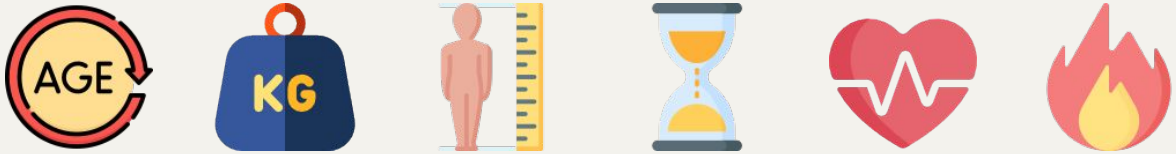
# Calorie Burn Prediction Model




**Team: Claudio & Vera**

# Project Overview

**Calorie Burn Prediction Model** that leverages key physiological and exercise-related factors

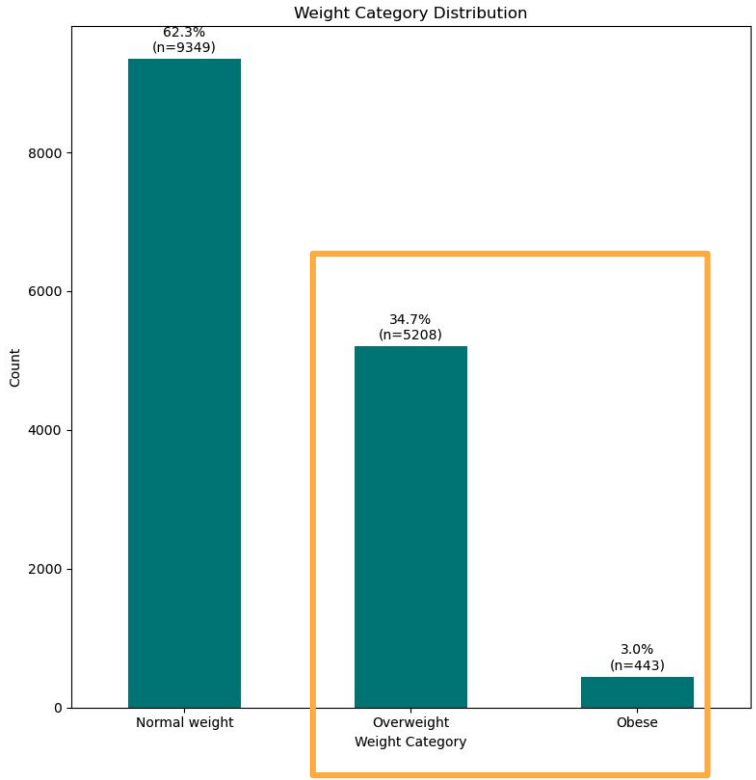


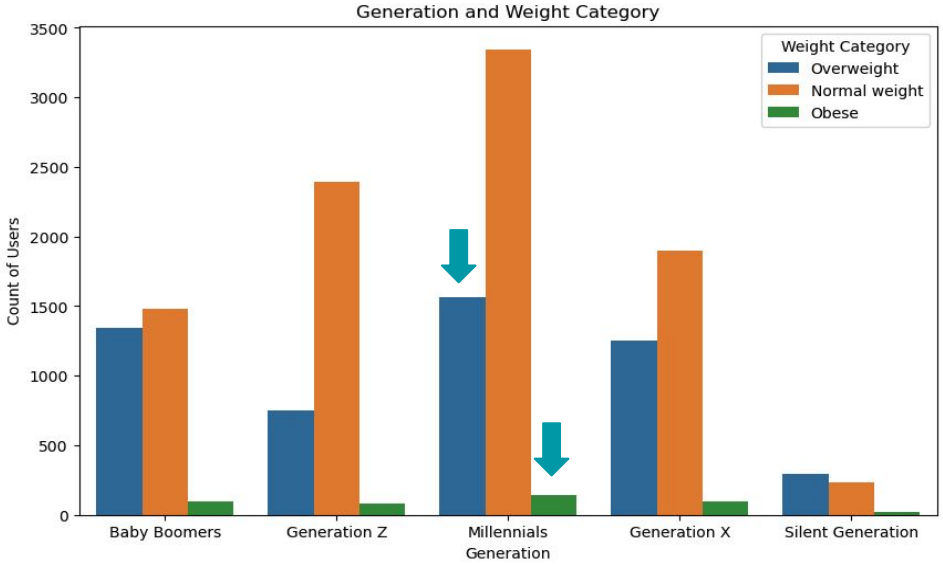

**Objective:** Enhance workout efficiency to maximize calorie burn results

**Potential Real-World Application:**

- Help users achieve **fitness goals efficiently** with optimized exercises
- Support **trainers and fitness apps** in delivering customized workout plans

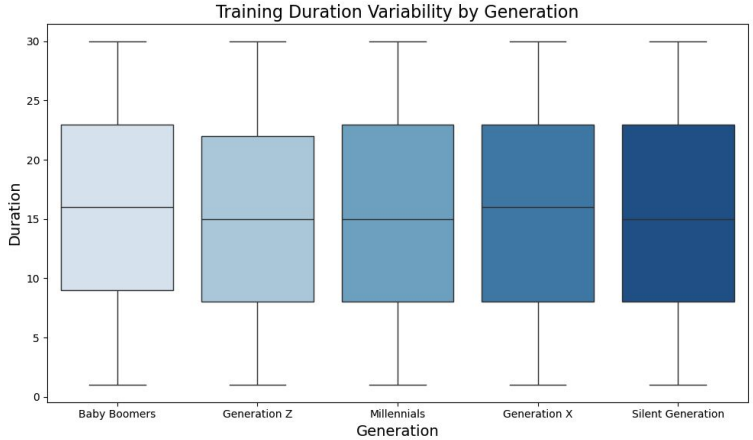
# Weight Insights

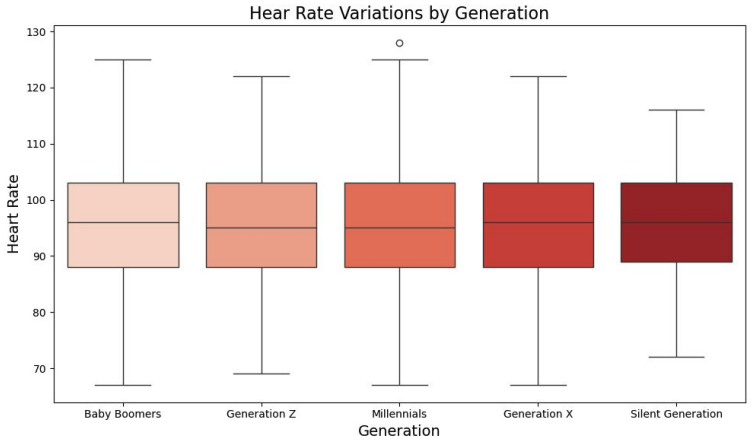


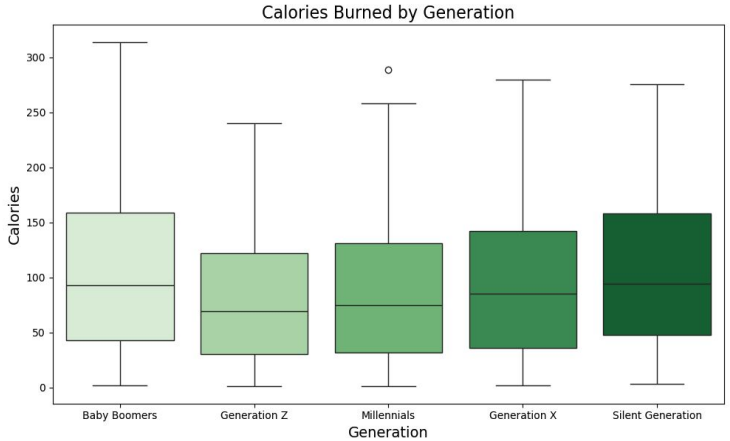




| Weight_Category | Normal weight | Obese | Overweight |
|---|---|---|---|
| Generation | | | |
| Millennials | 22.306667 | 0.933333 | 10.420000 |
| Baby Boomers | 9.853333 | 0.666667 | 8.973333 |
| Generation X | 12.680000 | 0.660000 | 8.366667 |
| Generation Z | 15.953333 | 0.540000 | 4.986667 |
| Silent Generation | 1.533333 | 0.153333 | 1.973333 |

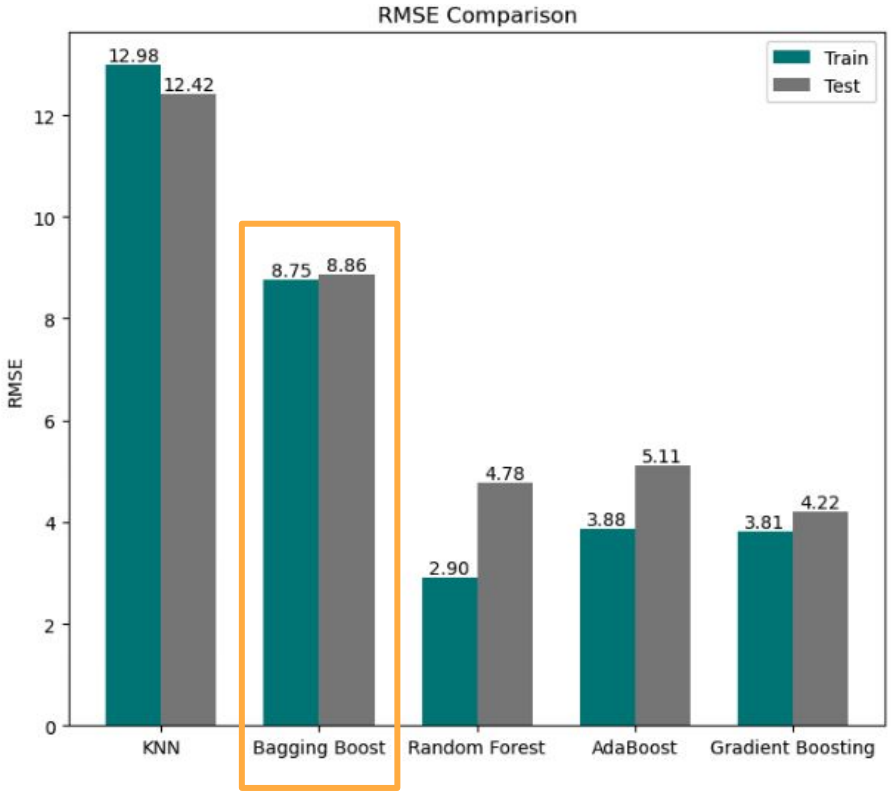
# Training Insights across Generations

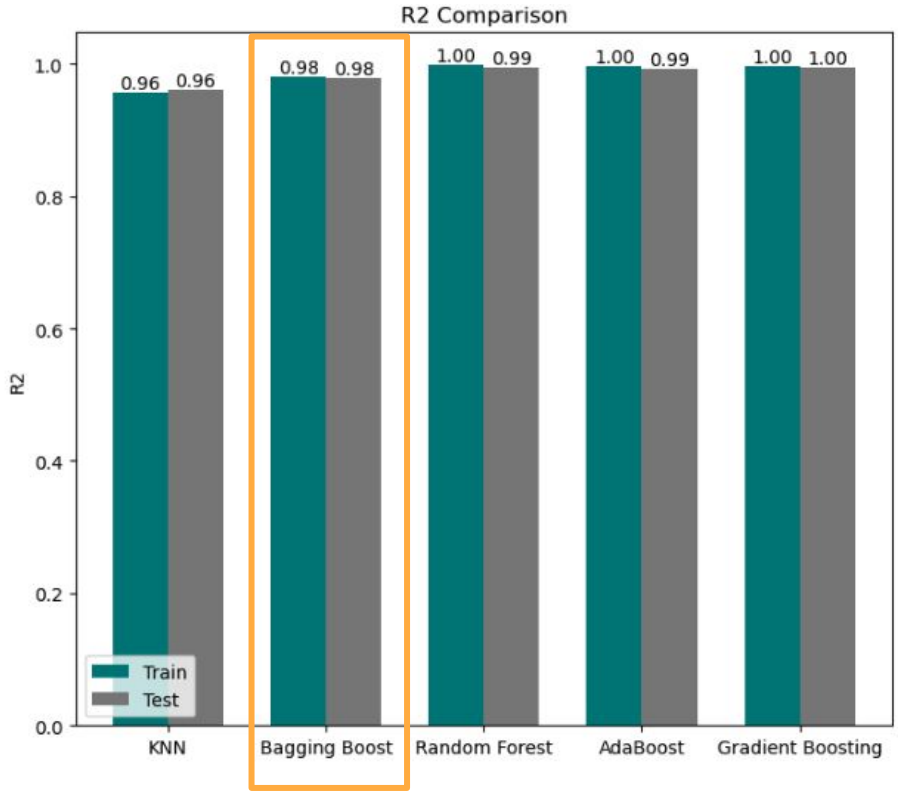


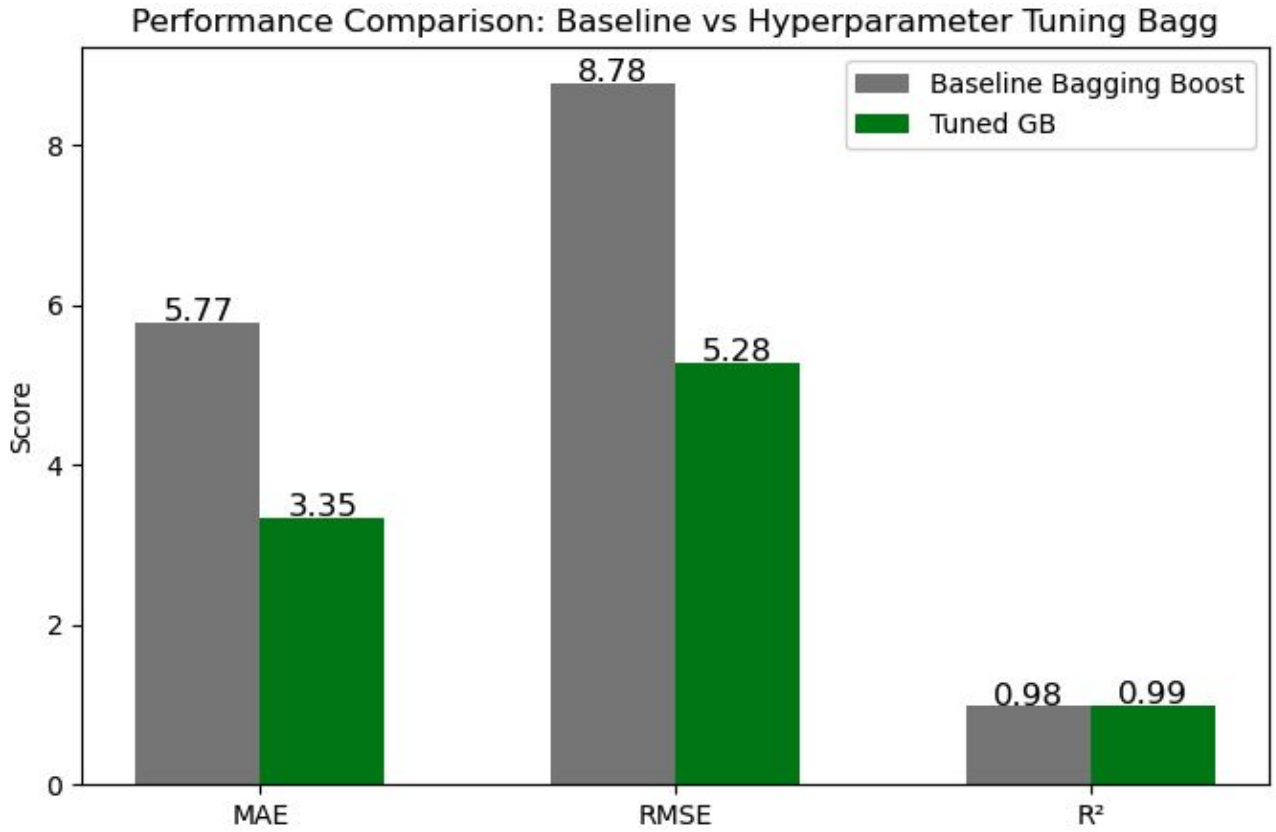



# Models Comparison



# HyperParameter Tuning - Bagging Boost

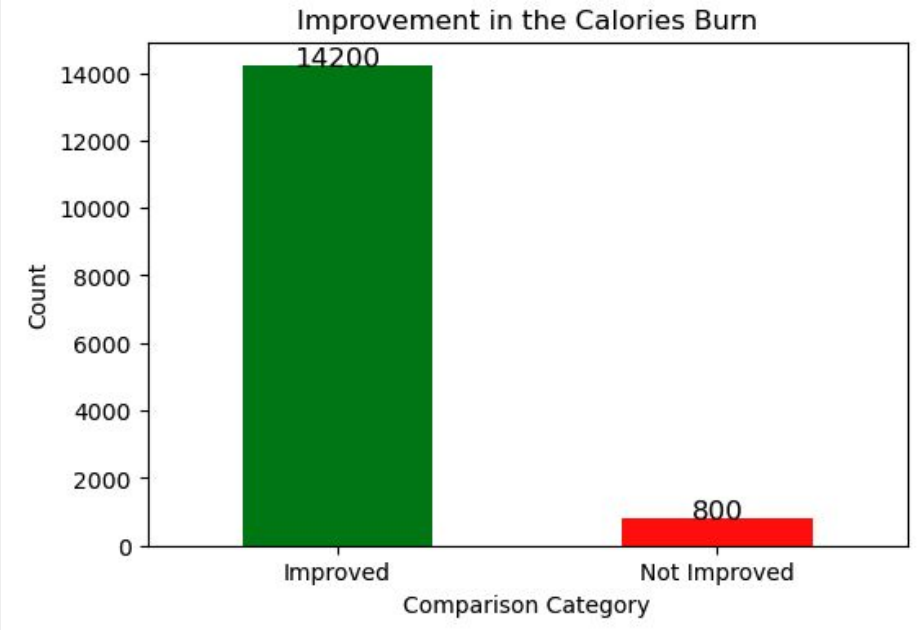


# Conclusions

This model provides **meaningful insights** into workout effectiveness, enabling users to optimize their sessions through **generation specific strategies** and **BPM-driven calorie burn optimization**.

It can be **powerful tool for personalized fitness guidance**, helping individuals achieve **more efficient and tailored workouts** for maximum results.

| | Generation | Common_Fitness_Trends | Estimated_BPM | Min_BPM | Max_BPM |
|---|---|---|---|---|---|
| 0 | Generation Alpha | Active play | 90–140 | 90 | 140 |
| 1 | Generation Z | HIIT | 134–190 | 134 | 190 |
| 2 | Millennials | Endurance sports | 114–162 | 114 | 162 |
| 3 | Generation X | Functional fitness | 95–133 | 95 | 133 |
| 4 | Baby Boomers | low-impact | 76–114 | 76 | 114 |
| 5 | Silent Generation | Gentle mobility | 60–100 | 60 | 100 |



# Challenges

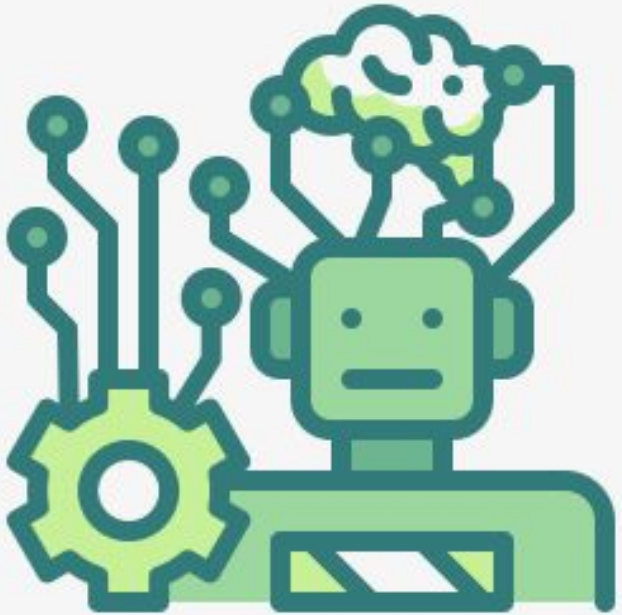
- Identify and Remove Irrelevant Features
- Choose the best parameters for Tuning
- Keep the notebook organized
- Time management

# Future Improvements

- Analyze the 800 cases where calorie burn did not improve, to identify potential model refinements
- Integrate workout duration as a key metric, optimizing BPM predictions to achieve the same calorie burn in shorter time
- Expand data inputs with body composition metrics, such as muscle mass vs. fat percentage, to improve the precision of calorie burn estimations

# Thank You!

Claudio & Vera



Calorie Burn Prediction Model - ML Project